

# PUNJAB DROUGHT-TOLERANT SEED VARIETIES

**Sector:** Agriculture

**Hazard addressed:** Droughts

**Targeted assets:** Sugarcane, Rice, Cotton, Wheat



**Description:** This measure promotes adoption of drought-tolerant and climate-resilient sugarcane seed varieties to improve productivity under water-stressed and heat-prone conditions. The intervention enhances crop survival, improves water-use efficiency, and stabilizes sugarcane yields during drought periods and irregular rainfall conditions while strengthening resilience of sugarcane farming systems.

**Priority Districts:** Lahore, Sahiwal, Pakpattan, Bahawalnagar, Vehari, Khanewal, Lodhran, Multan, Bahawalpur, Rahim Yar Khan, Rajanpur, Dera Ghazi Khan, and Muzaffargarh.

## TECHNICAL SPECIFICATIONS/DIMENSION

**Unit of implementation:** Hectare (ha) under drought-tolerant sugarcane cultivation.

**Scale assumptions:** Applicable in sugarcane-growing districts vulnerable to declining water availability, prolonged dry spells, and irrigation shortages.

**Key design parameters:** Use of certified drought-tolerant sugarcane seed varieties; compatibility with local agro-climatic conditions; timely seed distribution; farmer awareness; and climate-smart agronomic practices.

**Time horizon:** 2026 – 2032.

## IMPACT

**Risk reduction level:** High.

**Time to effectiveness:** Short- to medium-term following adoption during initial production cycles.

**Expected impact:** Improved yield stability under drought conditions, enhanced water-use efficiency, reduced crop losses during dry spells, and strengthened resilience of sugarcane farming systems.

## IMPLEMENTATION CONSIDERATIONS

**Enabling conditions:** Availability of certified seed supply chains, extension services, farmer awareness programs, and climate advisory systems.

**Required institutions:** Agriculture Department Punjab, Sugarcane Research Institutes, Seed Certification Agencies, Agricultural Extension Services, and development partners.

**Barriers:** Weak seed distribution systems, affordability constraints, limited technical awareness, and resistance to adoption of improved seed varieties.

## COST ASSUMPTIONS

**Unit cost:** Sugarcane: PKR 710 million (~USD 2.5 million), Wheat: PKR 365 million (~ USD 1.3 million), Rice: PKR 322 million (~ USD 1.2 million), Cotton: PKR 425 million (~ USD 1.5 million).

**Capital Expenditure (CAPEX):** USD 1.6 million in the first year (2026) then 1.5 million USD per year (2027 – 2032). Includes procurement and distribution of certified drought-tolerant sugarcane seed varieties, demonstration plots, extension services, and farmer awareness campaigns. (Sugarcane: USD 0.57 million, Wheat: USD 0.23 million, Rice: USD 0.35 million, Cotton: USD 0.43 million).

**Operational Expenditure (OPEX):** N/A

## CO-BENEFITS & TRADE-OFFS

**Environmental:** Reduced irrigation demand and improved efficiency of agricultural water use.

**Social:** Enhanced livelihood resilience and income stability for farming communities.

**Economic:** Reduced production losses during droughts and improved long-term profitability of crop cultivation.

**Potential risks:** Limited availability of certified planting material, variable adoption rates, and sensitivity to extreme climatic conditions.

## BENEFICIARIES

**Primary beneficiaries:** Farmers, agricultural cooperatives, extension departments, sugar industry stakeholders, and rural farming communities.

**Distributional aspects:** Particularly beneficial for irrigation-dependent sugarcane farmers vulnerable to drought stress and declining water availability.